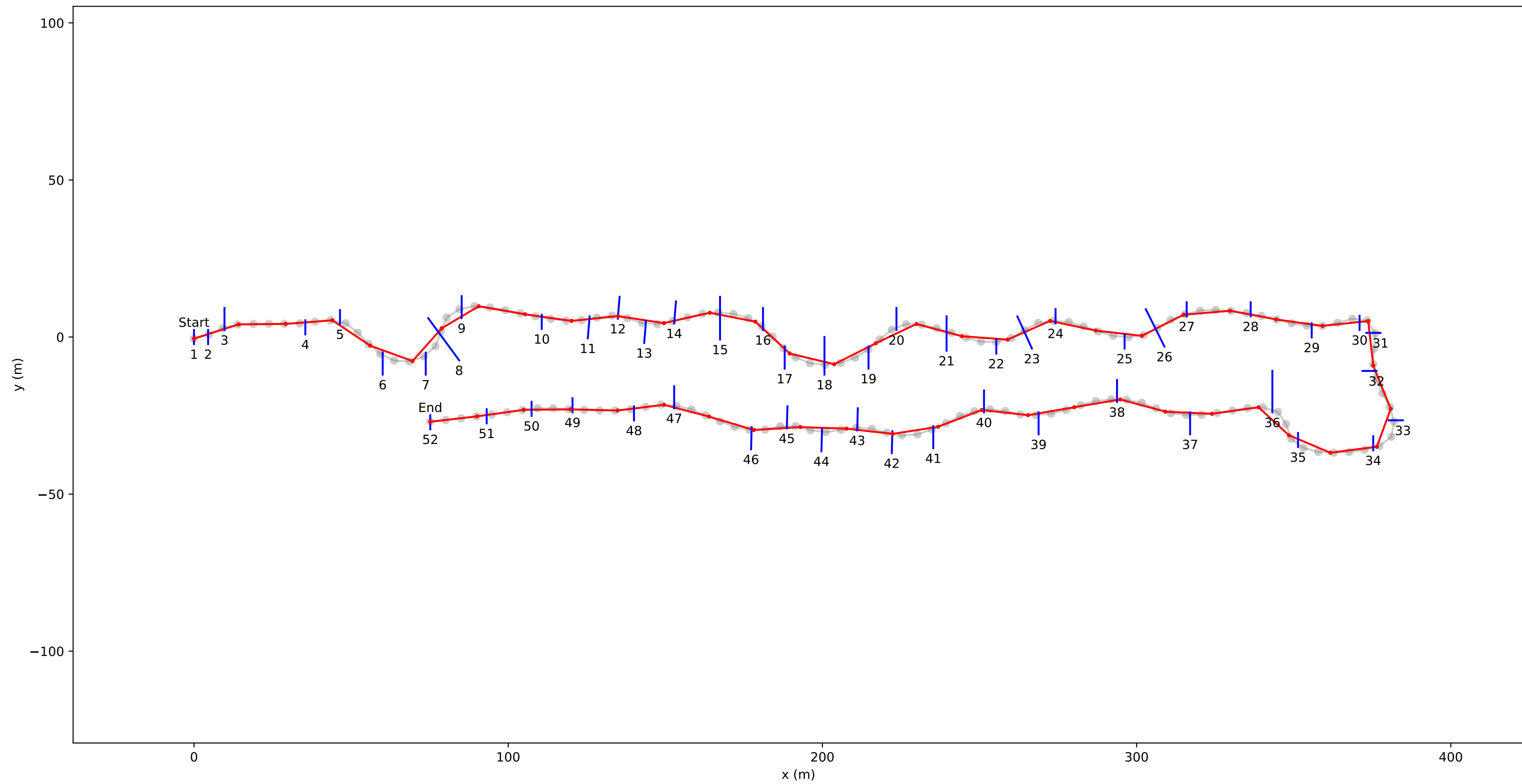


Map plot: optimized path through gates



Track Generation Report

=====
Generated: 2026-05-30 20:06:52
Gate file: ref_gates_autoX.csv
Gate count: 52
Saved DOFs file: opt_dofs_2026-05-30_20-06-52.npy

Optimization parameters

method: L-BFGS-B
maxiter: 500
ftol: 1e-07

Point counts

objective function points: 52
path generation points: 156

Results

total optimization time: 1865.210 s
total path length: 763.4142 m

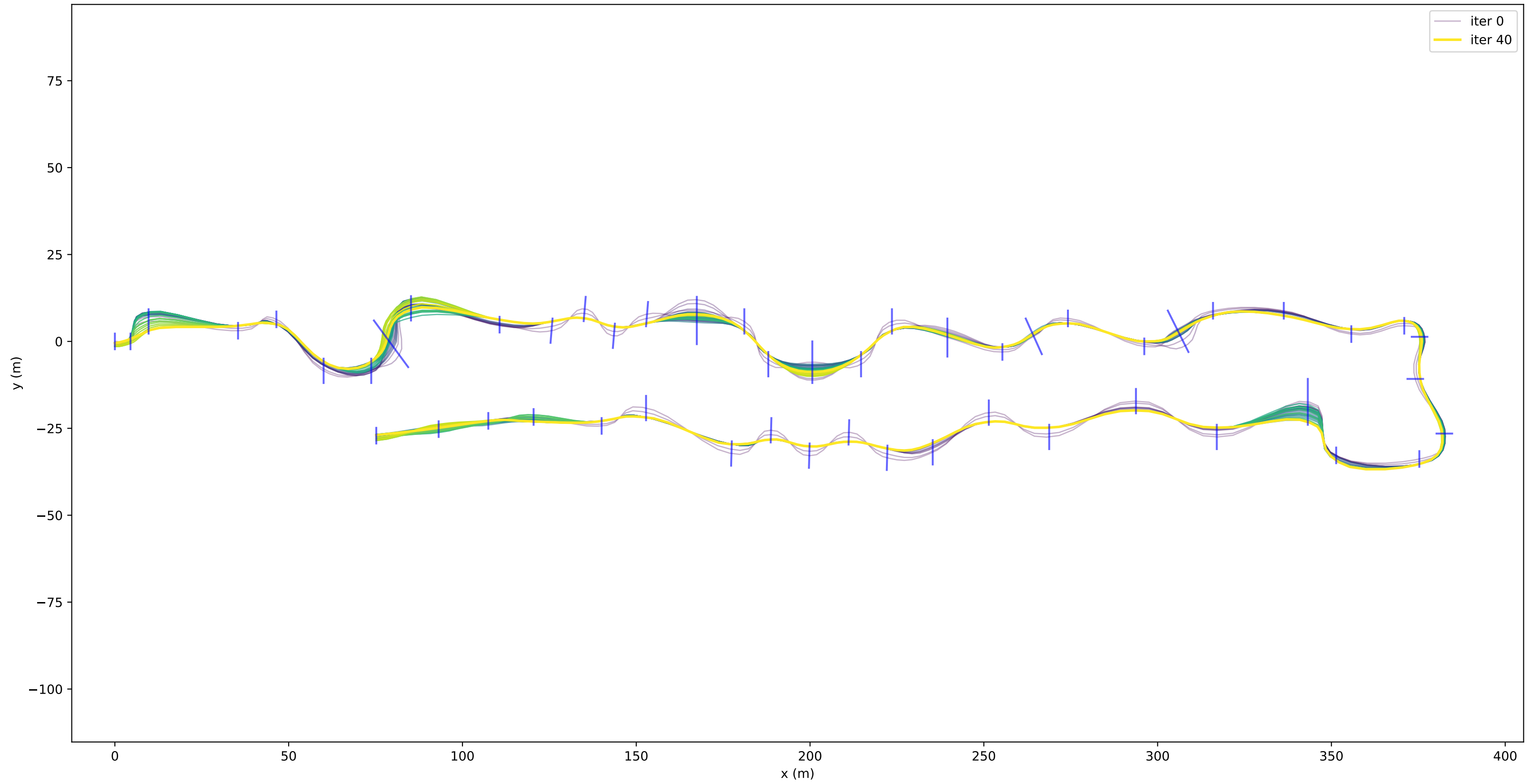
total objective function value: 2.275418

Per-step time estimate (one measured objective evaluation)

make spline from dofs: 12.48 ms
path length function: 0.61 ms
equidistant times generated: 0.61 ms
curvature objective function: 151.33 ms

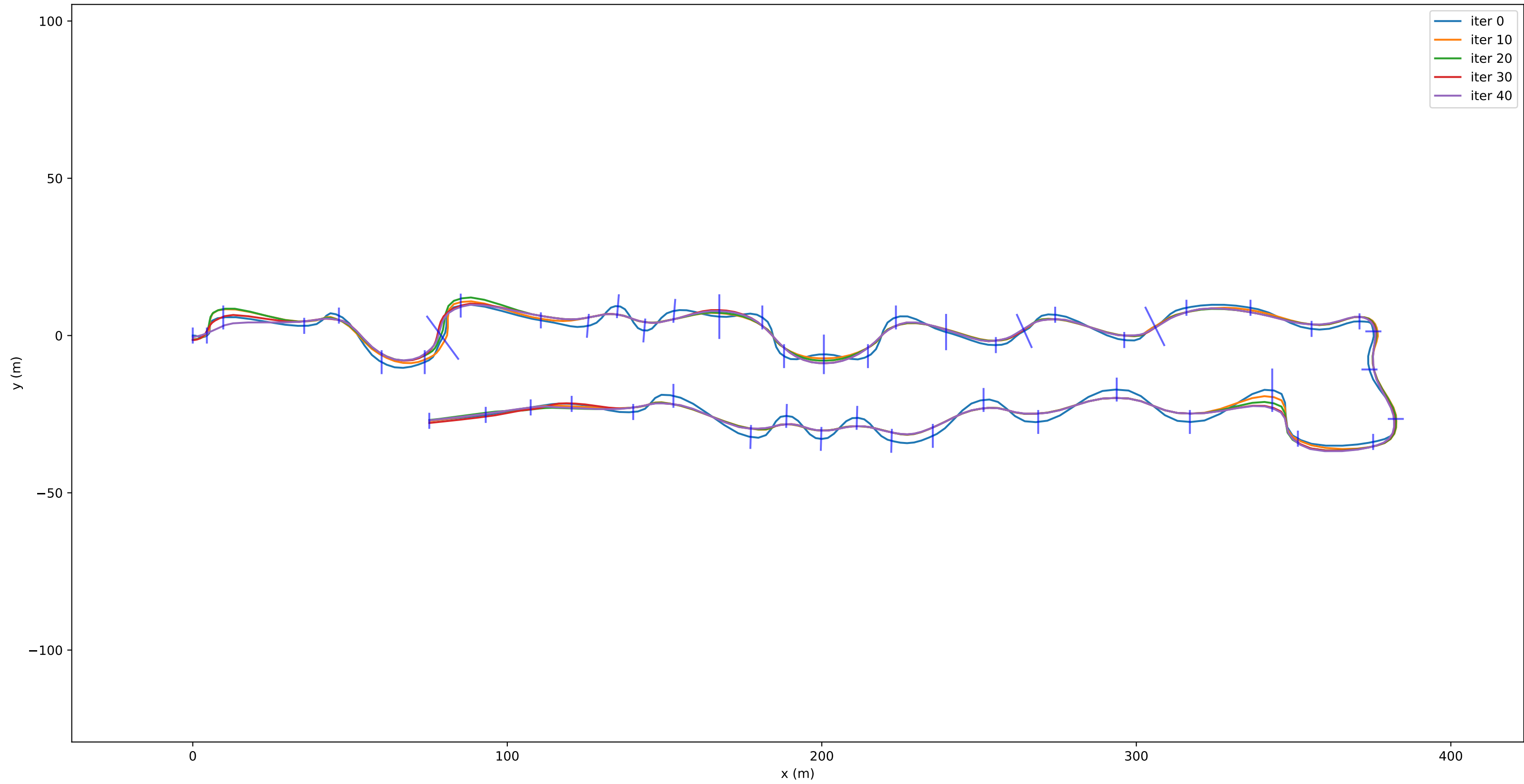
Optimization progression

Optimization progression (41 iterations)



Equally spaced iterations

5 equally spaced iterations



Instant radius vs distance

